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FINAL YEAR PROJECT PROPOSAL

Adaptive Explainable Ensemble with NLP Fusion for Pre-Launch Steam Game Reception Prediction under Varying Feature Availability

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1. Research Topic

The commercial video game industry has grown into one of the largest entertainment sectors globally, with Steam alone hosting over 90,000 titles as of 2025. For game developers and publishers, accurately anticipating how their game will be received by players before launch represents a significant unsolved challenge. A poor public reception, reflected in low user review scores, can severely impact sales, studio reputation, and future development capacity. Despite this, no evidence-based tool exists to assess audience reception risk during the design and production phase.

This project proposes an upgraded adaptive machine learning system that predicts whether a Steam game will achieve positive user reception defined as 75% or more positive reviews using only metadata available before launch. The original system trains specialist classifiers on graduated feature tiers and routes predictions to the most appropriate model based on which features are currently available. Following a structured cross-domain literature review across the film, mobile app, crowdfunding, and music prediction domains, the system has been enhanced with four evidence-backed improvements: derived feature engineering motivated by film industry research, a confidence-based routing mechanism grounded in NLP and healthcare prediction literature, a hybrid NLP and deep learning fusion tier using Sentence-BERT to encode developer-written game descriptions, and a dual SHAP and LIME explainability layer. These improvements are directly motivated by how other domains have solved structurally analogous pre-launch prediction problems.

The research topic is formally defined as follows:

Component	Definition	Applied to This Project
Task	What the system does	Binary classification: predict well-received (1) or not well-received (0) using only pre-launch metadata and developer-written descriptions
Domain	The problem area	Gaming analytics / Steam platform reception prediction
Constraint	The specific research angle	Pre-launch metadata and developer descriptions only; adaptive confidence-based multi-model routing under varying feature availability; NLP and deep learning fusion via Sentence-BERT; dual SHAP and LIME explainability at each planning stage
Motivation	Why this design	Cross-domain research across film, app store, crowdfunding, and music prediction demonstrates that these architectural patterns consistently improve prediction accuracy in analogous pre-launch classification tasks

This formulation specifies the task, domain, and key technical constraints that make the problem non-trivial and the contribution novel. The upgraded design is grounded in evidence from twelve cross-domain papers (CD1–CD12) and is scoped for implementation within three months.

2. Targeted Literature Review

A structured review of the existing literature was conducted in two phases. The first phase searched the gaming and machine learning domain using focused queries across Google Scholar, Semantic Scholar, arXiv, and the ACM Digital Library. Search terms included "Steam game reception prediction machine learning", "video game rating classification metadata features", "class imbalance ensemble tabular", and "SHAP explainability feature attribution". From an initial set of approximately twenty papers, ten were selected based on technical relevance, recency, and direct applicability to the proposed research gap. The second phase conducted a cross-domain literature search; motivated by the recognition that other domains have studied structurally analogous pre-launch prediction problems; using forward and backward snowballing on confirmed anchor papers across four domains.

The ten domain-specific papers are presented in the following matrix:

#	Authors & Year	Problem / Task	Dataset	Features Used	Method	Key Metrics	Primary Limitation
P1	Santos et al. (2019) ACM CSCW	Descriptive study of critic-user score discrepancies on Metacritic	1M+ Metacritic reviews	Review text, ratings, timestamps	Statistical analysis, linguistic analysis	Rating polarisation measures	Descriptive only — no predictive model; does not use pre-launch metadata
P2	Ma et al. (2025) Wiley IJCGT	Predict video game commercial revenue from platform data	Steam + Metacritic + Twitch (Sep 2024)	Peak CCU, follower count, total reviews, Twitch viewers, Metascore, price	CatBoost, Multiple Linear Regression	Feature importance, revenue correlation	All top features are post-launch; predict revenue not reception quality
P3	Al Tawil et al. (2025) iJIM Vol. 19	Classify Steam user review text by sentiment and feedback type	Steam user review text corpus	NLP text features from review content	ML classifiers with advanced feature extraction	Classification accuracy, F1	Post-hoc analysis of existing reviews; cannot predict reception before launch

P4	Pramanik (2025) arXiv 2601.11523	Predict video game review scores across platforms	Multi-platform game review data	Number of user reviews, gameplay duration, review patterns	Random Forest regression	Prediction accuracy, feature importance	Top predictor (review count) is post-launch; regression not binary classification
P5	Rahman et al. (2024) IJACSA Vol. 15	Predict number of Steam game players	Steam player activity data	Game features and player activity metrics	Random Forest, SVR, XGBoost with feature selection	Prediction accuracy, error metrics	Predicts player count not reception quality; uses post-launch engagement metrics
P6	Teja et al. (2023) E3S Web of Conferences	Predict Steam game rating score using regression	Steam game data	Game features and platform engagement metrics	Regression models	R2, prediction error	Regression not binary classification; includes post-launch metrics alongside metadata
P7	Hu, Wang and Xia (2024) DSAE Conference	Analyse Steam game popularity using machine learning	Steam game metadata and player engagement data	Game attributes, pricing, player activity metrics	Machine learning classifiers	Classification accuracy, popularity metrics	Predicts popularity (engagement) not user reception quality; uses post-launch player activity
P8	Mulla et al. (2025) JCST Vol. 15	Predict player churn in the gaming industry	Gaming platform user behaviour data	Player behavioural and activity features	Multiple ML algorithms comparative evaluation	Accuracy, F1, AUC	Targets player retention not game reception

							quality; uses post-launch behavioural signals unavailable before release
P9	Irawan et al. (2025) Scientific Reports	Mitigate class imbalance in binary classification (churn prediction)	Telecom customer churn data	Customer behavioural and account features	Ensemble methods + SMOTE comparison	F1, accuracy, AUC	Domain is telecom not gaming; imbalance ratios differ
P10	Lundberg and Lee (2017) NeurIPS Vol. 30	Unified framework for interpreting ML model predictions (SHAP)	Multiple benchmark datasets	All input features of trained models	SHAP (SHapley Additive exPlanations)	Feature importance attribution, theoretical consistency	Explains model after training — does not select features

2.1 Pattern Identified Across the Literature

A consistent pattern emerges across the eight domain papers (P1–P8). Every study that attempts to predict or explain game reception, success, or player behavior relies on metrics that are only available after a game has launched. Ma et al. (2025) use peak concurrent users, total review counts, and Twitch viewership, none of which exist before release. Pramanik (2025) identifies the number of user reviews as the most influential predictor, which is inherently a post-launch signal. Teja et al. (2023) incorporate post-launch engagement data alongside metadata features. Santos et al. (2019) and Al Tawil et al. (2025) both operate entirely on review text that only exists after players have experienced the game. Hu, Wang and Xia (2024) and Mulla et al. (2025) similarly use post-launch player activity and behavioral data as their primary input.

None of the reviewed papers restrict the prediction task to features available before launch. Furthermore, no study formulates the problem as a binary classification of reception quality on Steam, and no study examines which specific pre-launch characteristics are most predictive. Papers P9 and P10 establish the methodological toolkit, the imbalance handling strategies and SHAP-based explainability that this project applies to address these gaps.

2.2 Cross-Domain Literature Review

To identify how analogous pre-launch prediction problems have been solved in other domains, and to discover architectural patterns and methods that could improve this project, a structured cross-domain literature search was conducted. The search targeted four domains where a structurally similar prediction problem, classifying success or quality from pre-launch metadata before any user interaction has occurred, has been studied. The four domains are: film box office prediction, mobile app store success prediction, crowdfunding campaign success prediction, and music hit prediction. Forward snowballing (finding papers that cite a confirmed anchor) and backward snowballing (finding papers cited by an anchor) were applied to each confirmed paper. Twelve cross-domain papers were confirmed as directly relevant (CD1–CD12) and are presented below.

The cross-domain search was motivated by three questions:

(1) What features and methods have other domains found most effective for pre-launch binary classification?

(2) Have other domains integrated NLP or deep learning into structurally similar problems, and with what results?

(3) What architectural decisions, particularly around routing, explainability, and feature fusion have been validated by evidence from outside the gaming domain?

#	Authors & Year	Domain	Problem	Method	Key Finding & Implication for This Project
CD1	Sahu et al. (2023) Multimedia Tools and Applications DOI: 10.1007/s11042-022-13448-0	Film	Predict movie success at earliest production stage from pre-release metadata	K-fold Hybrid Deep Ensemble (KHDEM): DL models + ensemble classifiers + logistic regression combiner; achieved 96% accuracy	Derived feature engineering alone improved accuracy by 17.62% overusing raw features only. Directly motivates introducing interaction features (price-to-DLC ratio, genre concentration, platform coverage) in this project. Confirmed via backward snowball from MAMRP.
CD2	Bhavesh Kumar & Pande (2023) EAI Endorsed Transactions on Scalable Information Systems	Film	Explainable neural network for movie success prediction	Neural Network + SHAP for local and global feature attribution	Forward snowball from CD1. Extends the Sahu et al. architecture with SHAP explainability. Validates applying SHAP to hybrid DL+ensemble outputs,

#	Authors & Year	Domain	Problem	Method	Key Finding & Implication for This Project
					parallel to this project's dual explainability design.
CD3	Qin et al. / MAMRP (2023) Springer LNCS DOI: 10.1007/978-3-031-46664-9_44	Film	Multi-modal movie rating prediction combining structured metadata with transformer-based text features	Transformer-based extraction feeding into weight-based and tree-based fusion modules	Nearly 24% improvement over single-modality baseline. Validates hybrid fusion of transformer text features with structured metadata, the core architecture of Model E in this project.
CD4	Liao et al. (2022) Annals of Operations Research DOI: 10.1007/s10479-020-03804-4	Film	Early box office prediction using stacking ensemble	Stacking fusion model with meta-learner trained on base classifier probability outputs	Stacked meta-learner outperforms all individual classifiers and voting ensembles. Backward snowball for MAMRP. Provides evidence for optional Priority 5 stacked meta-learner across Models A–E.
CD5	Aleem & Noor (2024) Library Progress International, 44(3), pp. 1907–1918	App Store	Predict Android app success before Google Play launch from pre-launch app metadata	RF, SVM, XGBoost, KNN benchmarked on app category, content rating, price, size	Closest structural parallel to this project: app category = genre, content rating = age rating, price = price. Pre-launch constraint and feature types are identical. Independently validates the project's feature selection and problem formulation.
CD6	Gunduz (2024) PeerJ Computer Science DOI: 10.7717/peerj-cs.2316	Crowdfunding	Predict Kickstarter campaign success from pre-launch	BERT vs FastText representations + LSTM and GBM classifiers; BERT+LSTM	All text is pre-launch, written by the campaign creator before the campaign goes live, directly mirrors using Steam short_description

#	Authors & Year	Domain	Problem	Method	Key Finding & Implication for This Project
			campaign text (blurbs)	achieved accuracy 0.745	written by developers. BERT significantly outperforms FastText. Primary motivation for Sentence-BERT in Model E.
CD7	Maarouf, A., Feuerriegel, S. and Pröllochs, N. (2024) arXiv:2409.03668	Startup	Predict startup success by fusing LLM-encoded text descriptions with structured financial features	LLM text embeddings concatenated with structured features; SHAP applied across the full fused model	Demonstrates that developer-written text descriptions alongside structured features improve prediction. SHAP values aggregated across document embeddings provide unified explainability, blueprint for dual SHAP+LIME approach in this project.
CD8	Kasneci & Kasneci (2024) arXiv 2411.01645	General Tabular	Enrich tabular classification with contextual LLM embeddings across ensemble classifiers	RoBERTa and GPT-2 embeddings concatenated with structured tabular features; tested on RF, XGBoost, CatBoost across imbalanced datasets	LLM-derived features consistently rank among the most impactful in SHAP analysis. Gains most notable in XGBoost and CatBoost on imbalanced data. Directly validates using Sentence-BERT embeddings for Model E on this project's imbalanced dataset.
CD9	Gameiro, M.A. (2025) arXiv 2511.11673	Music / Lyrics	Synergistic fusion of Sentence-BERT embeddings with auxiliary structured features for lyrical classification	Synergistic Fusion Layer (SFL): gated mechanism modulating SBERT embeddings via structured features; Macro F1 0.9894 vs RF	SBERT + structured features via early fusion outperforms either modality alone. Closest architectural parallel to Model E. Shows structured features serve as effective contextual modulators for the text embedding.

#	Authors & Year	Domain	Problem	Method	Key Finding & Implication for This Project
				concatenation 0.9868	
CD10	Şencan et al. (2026) Applied Sciences DOI: 10.3390/app16094160	Social Media	Popularity prediction via multi-modal stacked ensemble	Logistic regression meta-learner on RF + XGBoost + Neural Network outputs; stratified 5-fold cross-validation to prevent leakage	Stacked generalisation with data-driven meta-learner dynamically weights specialist probabilities. Validates the optional Priority 5 stacked meta-learner for combining Models A–E output probabilities.
CD11	Alajmi, A. and Pergola, G. (2025/2026) arXiv 2512.23732	NLP Classification	Confidence-based routing for model selection in multi-model classification systems	Maximum class probability (predict_proba max) used as confidence score; inputs with low confidence escalated to more capable models	Defines and validates the confidence-based routing mechanism: route to the model expressing highest confidence rather than a fixed rule. Primary motivation for upgrading the rule-based router in this project.
CD12	Wang, J. et al. (2024) arXiv 2405.16413	Healthcare	Confidence-driven model selection for EHR-based risk prediction	Average predict_proba as confidence criterion; threshold sigma determines whether prediction is confident or should escalate	Second independent source validating confidence-based routing from a completely different domain. Cross-validates CD11 and strengthens the case for replacing the rule-based router.

2.3 Patterns and Improvement Opportunities Identified from Cross-Domain Research

Analysis of CD1-CD12 across four domains reveals six consistent findings, each with concrete implications for upgrading the current project architecture. These findings form the evidence base for the five proposed improvements described in Section 4.

Finding 1: Derived Feature Engineering Gives Disproportionate Performance Gains

Source domain: Film industry. Sahu et al. (CD1) demonstrate that introducing interaction features computed from existing raw metadata before any model change, improved early-stage success prediction accuracy by 17.62%. The study confirms that raw features alone underutilize the signal available in the data, and that simple mathematical relationships between features capture patterns that individual raw features cannot. Forward snowballing from CD1 led to Bhavesh Kumar and Pande (CD2), who extend this finding with SHAP explainability on the derived feature outputs.

Implication: This project currently uses only raw features: price, DLC count, and age rating with no interaction terms. Deriving features such as a price-to-DLC ratio (monetization aggression index), a genre concentration score, and a platform coverage ratio is expected to produce meaningful accuracy improvements at negligible additional implementation cost.

Finding 2: Pre-Launch App Success Prediction is a Direct Structural Parallel

Source domain: Mobile app stores. Aleem and Noor (CD5) predict Android app success before launching using RF, SVM, XGBoost, and KNN on pre-launch app metadata. App category maps exactly to game genre; content rating maps to age rating; price maps to price. The pre-launch constraint, no post-launch signals permitted, is identical to this project. Their best-performing model (SVM at 80.34% accuracy) and the feature set used provide an independent validation of this project's core design decisions from an entirely separate domain.

Implication: The cross-domain parallel confirms that the problem is solvable with the feature types already used in this project. It also shows that the comparison of RF, XGBoost, and SVM across the same feature types is a natural evaluation design to adopt.

Finding 3: Hybrid NLP Fusion of Pre-Launch Text with Structured Features Consistently Improves Accuracy

Source domain: Crowdfunding, startup prediction, general tabular ML, and music classification. Three papers from distinct domains converge on the same finding. Gunduz (CD6) predicts Kickstarter success from pre-launch campaign blurbs, text written by the campaign creator before the campaign goes live, finding that BERT representations outperform FastText with BERT+LSTM achieving 0.745 accuracy. Kasneci and Kasneci (CD8) demonstrate that LLM embeddings integrated with structured tabular features improve XGBoost and CatBoost performance on imbalanced datasets, with LLM-derived features ranking among the most impactful in SHAP analysis. Gameiro (CD9) validates a gated Sentence-BERT fusion architecture for lyrical classification, achieving Macro F1 of 0.9894 against a concatenation baseline of 0.9868.

Maarouf et al. (CD7) additionally demonstrate that SHAP can be applied across the full fused model including text embedding dimensions, providing a unified explainability framework that covers both modalities. Steam’s short_description field, written by the game developer as part of the store page before launch is directly analogous to the pre-launch text used in all three studies. Coverage in the filtered dataset is 99.8% (only 37 games missing).

Implication: The consistent cross-domain evidence justifies adding Model E as a fifth specialist model, using Sentence-BERT to encode the short_description and fusing the resulting embedding with structured features. The SBERT approach is preferred over fine-tuned BERT due to the computational constraints of Google Colab and the 3-month implementation timeline. LIME is added to provide word-level attribution for the text portion of Model E.

Finding 4: Confidence-Based Routing Outperforms Rule-Based Model Selection

Source domain: NLP classification and healthcare prediction. Two papers from independent domains demonstrate that routing to the model expressing the highest prediction confidence outperforms selecting models by rule. Alajmi and Pergola (CD11) define confidence as the maximum class probability from predict_proba(): inputs where this value meets a threshold are accepted; inputs where it does not are escalated to a more capable model. Wang et al. (CD12) apply the same principle in Alzheimer's disease risk prediction from electronic health records, confirming the pattern generalizes across domains.

Implication: The current rule-based router selects the richest feature tier that is fully satisfied, which means it always routes to Model B if Tier 2 features are available, even if Model B is only 52% confident and Model A is 89% confident. Replacing this with a confidence-based router that queries each eligible specialists predict_proba() output and routes to the most confident model is a small implementation change with measurable accuracy and architectural improvements.

Finding 5: Stacked Meta-Learner Further Improves Multi-Model Ensemble Performance

Source domain: Film box office and social media popularity prediction. Liao et al. (CD4) demonstrate that a stacking fusion model trained in the probability outputs of base classifiers outperforms both individual classifiers and simple voting ensembles for early box office prediction. Şencan et al. (CD10) extends this with a logistic regression meta-learner trained via stratified 5-fold cross-validation on the outputs of RF, XGBoost, and Neural Network specialists, preventing data leakage while dynamically weighing specialist contributions.

Implication: A stacked meta-learner taking the probability outputs of Models A-E as inputs and learning optimal combination weights is a natural Priority 5 extension. This is scoped as optional within the 3-month timeline, to be implemented if Priorities 1-4 are complete.

Finding 6: Multimodal Fusion Architecture is Validated Across Multiple Independent Domains

Source domain: Film ratings and music/lyrical classification. Qin et al. (CD3) and Gameiro (CD9) from two entirely separate domains both validate that combining transformer-based text features with structured metadata outperforms either modality alone, with CD3 showing a 24% gain over single-modality baselines and CD9 showing a Macro F1 improvement of 0.9894 versus 0.9868. This cross-domain convergence provides strong evidence that the hybrid architecture of Model E is methodologically sound.

Implication: The consistency of this finding across film, crowdfunding, startup, music, and general tabular domains, five independent research lines, provides strong justification for the Model E design and addresses any examiner concern about whether NLP fusion is appropriate for this specific task.

3. Research Gap

The literature review reveals a clear and specific gap: while several studies have explored game success prediction and reception analysis, none has examined whether pre-launch game metadata alone like the genre, pricing model, DLC count, content tags, and age rating is sufficient to predict whether a game will be well-received by Steam players, and none has systematically evaluated which of these pre-launch characteristics are the strongest predictors of user reception.

This gap is not merely a matter of missing accuracy improvements. It represents a fundamentally different research question. Existing work is reactive; it analyses what happened after a game launched. The proposed project is predictive, it asks what can be inferred from decisions the developer makes during production, before any player has encountered the game. The practical consequence of this gap is that game developers currently lack any evidence-based tool to assess audience reception risk at the point where it would be most useful: during the design, pricing, and content planning phase.

The gap is further supported by the validated absence of directly comparable work. A targeted search using the Consensus AI academic search tool returned no papers meeting all three criteria simultaneously: prediction of user reception quality, using game-level metadata features, at or before the time of launch without relying on review text or post-launch engagement signals. This confirms that the proposed task formulation is novel within the current body of literature.

The gap can be characterized as a combination of four specific deficits. The method gap lies in the absence of a binary classification approach for pre-launch reception prediction. The data gap reflects the lack of any labelled benchmark combining pre-launch game metadata with binary reception outcomes on Steam. The evaluation gap stems from the absence of systematic feature ablation studies that quantify the marginal contribution of genre, tags, and pricing to reception predictability. The deployment gap is the deepest structural limitation: every existing study assumes a complete, fixed set of features is available at the point of prediction. Developers know genre and price at concept stage, DLC structure during production planning, and content tags closer to release. No existing system produces stage-appropriate risk assessments under this progressively expanding feature availability. Steininger et al. (2022) and related work on missing-feature handling in tabular ML address imputation strategies but not graduated, domain-structured feature availability.

The cross-domain review in Section 2.2 and 2.3 identifies two additional gaps confirmed by cross-domain evidence. The description gap: no study in the gaming domain has assessed whether developer-written pre-launch text descriptions carry additional reception signal beyond structured metadata. In three independent analogous domains: crowdfunding (Gunduz, 2024), general tabular ML (Kasneci and Kasneci, 2024), and music classification (Gameiro, 2025), this question has been studied and confirmed affirmatively. The Steam short_description field provides a direct equivalent for this investigation. The routing gap: no adaptive gaming prediction system uses confidence-based routing to dynamically select specialist models at inference time. CD11 and CD12 from independent domains demonstrate this is a superior mechanism to rule-based selection, and no study in the gaming or tabular ML domain has applied it to a tiered specialist architecture of this type.

4. Contribution to the Body of Knowledge

This project makes nine contributions to the body of knowledge, combining the original four contributions with five additional contributions motivated directly by the cross-domain research findings.

Contribution 1: Adaptive Multi-Model Ensemble with Feature-Tier Routing. The primary technical contribution is a system training five specialist classifiers on graduated feature tiers corresponding to stages of game development. Model A (Core) trains on price, genre, and age rating. Model B (Monetization) adds DLC count and pricing model. Model C (Content) adds content tags and supported languages. Model D (Full Tabular) trains on all structured features. Model E (Description) fuses Sentence-BERT-encoded game descriptions with structured features. At inference time, a confidence-based router selects the specialist expressing the highest prediction confidence from among those eligible for the available features. This enables progressive prediction developers to receive meaningful risk assessments at any planning stage.

Contribution 2: Systematic Ensemble Benchmarking. Comparative benchmarking of Logistic Regression, Random Forest, XGBoost, and CatBoost under three class imbalance handling strategies across the full structured feature set produces the first systematic benchmark of ensemble methods for pre-launch Steam game reception prediction. A missing-feature ablation study quantifies performance degradation as features are withheld and the recovery provided by tier-matched specialist models.

Contribution 3: Labelled Pre-Launch Reception Dataset. A filtered, labelled dataset of 20,383 Steam games with binary reception labels is produced as a reusable research artefact, the first benchmark of this type for the gaming domain.

Contribution 4: Class Imbalance Finding in the Gaming Domain. The finding that for a 72%/28% class split, no imbalance correction strategy outperforms SMOTE across the ensemble configurations tested, extends the class imbalance literature into the game reception prediction domain.

Contribution 5: Derived Feature Engineering for Pre-Launch Game Metadata. Motivated by Sahu et al. (CD1) demonstrating 17.62% accuracy improvement from derived interaction features in an analogous early-stage film prediction domain, this project introduces three derived features computed from existing raw inputs: a price-to-DLC ratio (monetization aggression index), a genre concentration score, and a platform coverage ratio. These signals capture relationships absent from raw inputs and are applied across all five specialist models at zero additional data collection cost.

Contribution 6: Confidence-Based Adaptive Router. The rule-based router is upgraded to a confidence-based mechanism motivated motivated by Alajmi and Pergola (CD11) and Wang et al. (CD12). Rather than selecting the richest satisfied feature tier by rule, the router queries each eligible specialists predict_proba() output and routes to the model expressing the highest-class probability. This directly addresses the routing gap identified in Section 3 and is validated by two independently sourced cross-domain papers from distinct domains.

Contribution 7: Model E: NLP and Deep Learning Hybrid Fusion Tier. A fifth specialist model adds a Sentence-BERT transformer encoder (all-MiniLM-L6-v2) as the deep learning component. The 384-dimensional SBERT embeddings are reduced to 50 dimensions via PCA fitted on the training split and concatenated with 36 structured features to produce an 86-dimensional fused matrix. This directly addresses the description gap in Section 3 and is motivated by Gunduz (CD6), Kasneci and Kasneci (CD8), and Gameiro (CD9), three independently sourced cross-domain papers confirming that pre-launch text fusion improves classification accuracy. Coverage of the short_description field is 99.8% in the filtered dataset, with 37 missing games automatically routing to Model D.

Contribution 8: Dual SHAP and LIME Explainability. The explainability layer is extended to a dual setup motivated by Maarouf et al. (CD7). SHAP provides feature attribution for structured features across all five specialist models, generating tier-specific attribution analyses that explain which features are driving the prediction at each planning stage. LIME provides word-level attribution for the text portion of Model E, identifying which words in the developer's game description contributed most to the prediction. Together they provide complete, modality-specific explainability across the entire pipeline.

Contribution 9: Cross-Domain Validated Architecture. The enhanced system is the first in the gaming domain to be explicitly designed and validated through evidence from analogous implementations in film (CD1–CD4), app stores (CD5), crowdfunding (CD6–CD7), general tabular ML (CD8), music (CD9), social media (CD10), NLP classification (CD11), and healthcare (CD12). Each major architectural decision is supported by at least two independently sourced cross-domain papers, providing substantially stronger methodological grounding than domain-internal validation alone.

The five improvements derived from cross-domain research are summarized in the following priority table, ordered by implementation effort and evidence strength:

Priority	Improvement	Effort	Primary Cross-Domain Sources	Expected Outcome
1	Derived Feature Engineering	Low	Sahu et al. (2023) CD1	17.62% accuracy gain in analogous film domain from derived features; applies to all tiers with no new data required
2	Confidence-Based Router	Moderate	Alajmi and Pergola (2025/2026): CD11, Wang et al. (2024): CD12	Dynamic routing to most confident specialist replaces hard tier rules; validated by two independent cross-domain papers
3	Model E: SBERT Hybrid Fusion (NLP + DL)	Moderate	Gunduz 2024: CD6, Kasneci & Kasneci 2024: CD8, Gameiro 2025: CD9, Maarouf et al. 2024: CD7	Tier 5 added; satisfies NLP and DL requirements; 99.8% dataset coverage; three independent domains confirm improvement
4	SHAP + LIME Dual Explainability	Low–Moderate	Maarouf et al. (2024): CD7, Lundberg & Lee (2017)	Complete modality-specific XAI: SHAP for structured features, LIME for text in Model E
5 (Optional)	Stacked Meta-Learner	Moderate–High	Liao et al. 2022: CD4, Şencan et al. 2026: CD10	Logistic regression meta-learner on specialist probability outputs; implement only if Priorities 1–4 are complete within 3-month window

5. Feature Set Identification and Justification

The primary data modality for this project is tabular, with features derived from structured game metadata available on the Steam platform before a game launches. Based on the original feature analysis and the cross-domain research in Section 2.3, features are now organized into five groups: raw numerical features, derived numerical features (new), genre-based categorical features, tag-based categorical features, and text features (new). The first four groups form the input space for Models A-D. All five groups are used by Model E.

The raw numerical features are price in USD, DLC count, and required age rating. Price is the most consequential feature, contributing 20.6% of the model's predictive power, more than twice the second feature, because higher-priced games are held to commensurately higher standards by players. DLC count contributes 8.2% of importance, reflecting gaming community sensitivity to monetization structure. Age rating provides a proxy for content maturity and target audience specificity.

Genre features are constructed by one-hot encoding the top ten genres from the Steam genres field, which developers assign during store page setup. Genre encoding improved model F1 by 12.7 points above raw features alone. The Adventure genre (importance 0.031) and Indie genre (importance 0.027) show the highest individual predictive power.

Tag features are constructed by one-hot encoding the top twenty tags from the Steam user tag field. Tag encoding improved F1 by a further 7.0 points above genres alone. The Singleplayer (0.031), 2D (0.030), and Atmospheric (0.028) tags show the highest individual importance. A known limitation is that while genre-type tags are established by developers during store setup, some user-applied tags may accumulate post-launch. The features selected are predominantly genre-type descriptors, and this limitation is documented as future work.

Feature Group	Feature	Source	Importance	Justification
Raw numerical	Price (USD)	Steam price field	0.206 (highest)	Higher-priced games face greater scrutiny; price-to-value gap is the dominant predictor
Raw numerical	DLC count	Steam DLC listings	0.082 (second)	Number of paid add-on packs signals monetization aggression; highly community-sensitive
Raw numerical	Required age rating	Steam age field	Low	Proxy for content maturity and target audience specificity
Genre features	Top 10 genres (one-hot)	Steam genres field	Up to 0.031 per genre	Genre encoding adds +12.7 F1 points; set by developer before launch; Adventure and Indie most predictive
Tag features	Top 20 tags (one-hot)	Steam user tags field	Up to 0.031 per tag	Tag encoding adds +7.0 F1 points over genres alone; Singleplayer, 2D, Atmospheric most predictive

Following the cross-domain research findings, specifically Finding 1 from Sahu et al. (CD1) demonstrating 17.62% accuracy improvement from derived interaction features, three derived features and one text feature are introduced:

Feature Group	Feature	Computation	Motivation	Justification
Derived numerical	Price-to-DLC ratio	price / (dlc_count + 1)	Sahu et al. (CD1): derived features improve accuracy by 17.62% in analogous domain	Monetization aggression index: high ratio signals premium pricing relative to content volume; a pattern associated with community backlash
Derived numerical	Genre concentration score	genres_assigned / max_possible_genres	Sahu et al. (CD1): interaction features between raw inputs carry signal not in individual features	Specificity of categorization: precisely targeted games signal clearer audience expectations and tighter quality standards
Derived numerical	Platform coverage ratio	(Windows + Mac + Linux) / 3	Sahu et al. (CD1): all derived features applied across all specialist tiers at no additional cost	Proportion of supported platforms reflects development investment breadth and commitment to the game
Text (Model E only)	short_description	SBERT (all-MiniLM-L6-v2) → 384-dim → PCA 50-dim → concatenated with structured features	Gunduz (CD6), Kasneci & Kasneci (CD8), Gameiro (CD9): pre-launch text fusion improves	Developer-written pre-launch description text; 99.8% coverage in filtered dataset (37 games missing); captures marketing intent and game identity signals absent from structured metadata

Feature Group	Feature	Computation	Motivation	Justification
			accuracy in 3 independent domains	

5.1 Feature Tier Grouping for Adaptive Architecture

Features are organized into five graduated tiers corresponding to stages of the game development process. Each tier maps to a specialist classifier trained exclusively on features available at that stage. The tier structure is updated from the original four tiers to five, with derived features applied across all tiers and Model E added as Tier 5.

Tier	Planning Stage	Features Included	Feature Count	Specialist Model
Core (Tier 1)	Concept stage: earliest design decisions	Price, Genre (top 10 one-hot), Age rating, Genre concentration score (derived)	14	Model A: trained on Core features only
Monetisation (Tier 2)	Production planning: business model decided	Core + DLC count, Free-to-play flag, In-App Purchases tag, Price-to-DLC ratio (derived)	18	Model B: trained on Core + Monetization features
Content (Tier 3)	Content finalization: creative decisions locked	Core + Top content tags, supported languages count, Demo flag, Platform coverage ratio (derived)	33	Model C: trained on Core + Content features
Full (Tier 4)	Pre-launch: all metadata submitted to Steam	All structured features from Tiers 1–3 combined with all 3 derived features (36 features total)	36	Model D: full tabular specialist, current best baseline (F1 0.822)
Description (Tier 5)	Pre-launch: developer description also available	All Tier 4 features + SBERT embedding of short_description reduced to 50 PCA dimensions	86	Model E: hybrid NLP and DL fusion specialist, new addition from cross-domain research

The router logic is upgraded from rule-based to confidence-based selection, motivated by Alajmi and Pergola (CD11) and Wang et al. (CD12). At inference time, the system identifies all feature tiers fully satisfied by the developer's input and queries each eligible specialist's predict_proba() output. The router selects the specialist expressing the highest-class probability. If short_description is available and non-empty, Model E is included in the eligible set alongside Model D. If only Core features are provided, Model A is used directly. This ensures the prediction is always made by the most confident available specialist, addressing the routing gap identified in Section 3 while retaining the tier-based fallback behavior that avoids inputting missing features into a model trained on complete data.

6. Dataset Identification and Evaluation

Five candidate datasets were identified and systematically evaluated against the requirements of the proposed task. The primary requirement is that a dataset must contain both pre-launch game metadata features and a reliable measure of user reception quality.

Dataset	Source	Size	Label?	Pre-launch Features?	Decision
Steam Games Dataset 2025	Kaggle: artermiloff (Mar 2025)	89,618 games	Yes: pct_pos_total	Yes: genres, tags, price, DLC, age rating, short_description (99.8% coverage)	PRIMARY DATASET selected. Contains all required structured features and short_description for Model E at 99.8% coverage after filtering.
Steam Multi-Modal 2025	Kaggle: crainbramp (2025)	239,000+ games	Yes, review ratings	Yes, extensive metadata	Excluded: SQL/embedding structure requires specialist setup beyond project scope
Metacritic Games Scraper 2025	Kaggle: zaireali (Feb 2025)	~13,000 games	Derived, critic vs user gap	Partial — limited metadata richness	Excluded: insufficient pre-launch feature coverage for full feature set and no description field
Steam + Reviews 2020–2024	Mendeley Data (Jun 2025)	23,000+ games	Yes, ratings present	Yes, metadata + review text	Excluded: review text focus introduces post-launch dependency; complexity beyond scope
Metacritic Best Games 2025	Kaggle: davutb (Jan 2026)	13,000+ games	Yes, scores present	Partial	Excluded: selection bias toward high-scoring games would underrepresent the not-well-received class

The Steam Games Dataset 2025, maintained by artermiloff on Kaggle and updated as of March 2025 using the Steam API and SteamSpy, was selected as the primary dataset. It contains 89,618 game entries with genre, category, tag, pricing, DLC, age rating, short_description, and review percentage fields. After filtering to games with 100 or more total reviews, ensuring the reception label is stable and not driven by a small number of early reviews, the working dataset comprises 20,383 games. Of these, 14,631 (71.8%) are well-received (75% or more positive reviews) and 5,752 (28.2%) are not well-received. The 72%/28% class distribution represents a mild imbalance addressed through comparative evaluation of multiple handling strategies. Of the 20,383 filtered games, 20,346 (99.8%) have non-empty short_description after HTML stripping, with only 37 games missing this field. Those 37 games fall back automatically to Model D in the confidence-based router.

7. Identified Challenges

Seven realistic technical and practical challenges are anticipated in the execution of this upgraded project.

The first challenge concerns class imbalance. With 71.8% of games well-received and 28.2% not well-received, a naive classifier could achieve over 70% accuracy by always predicting the majority class. This makes accuracy an unreliable primary metric. The project addresses this by using F1 score on the minority class as the primary performance measure and by systematically comparing three imbalance handling strategies: SMOTE, class weighting, and no correction.

The second challenge is the pre-launch validity of tag features. Steam tags are applied by users after launch, meaning some tags may technically constitute post-launch information. This is mitigated by restricting tag selection to genre-type descriptors established before launch, with this assumption documented as a limitation.

The third challenge involves the label threshold definition. The 75% positive review threshold is a modelling decision rather than an industry standard. This is addressed through sensitivity analysis, and the threshold is justified by its alignment with Steam's own "Mostly Positive" review category boundary.

The fourth challenge is genre and tag feature overlap. Steam's genres and tags fields sometimes include the same descriptors such as "Indie" or "Action", potentially introducing multicollinearity. Both sources are retained because they represent different annotation processes. The model's stable cross-validation performance (standard deviation of 0.002 across five folds) confirms this does not cause overfitting.

The fifth challenge is the risk of overfitting on binary one-hot features. With 36 structured features, many binaries, the model must learn sparse patterns. This is addressed through stratified five-fold cross-validation across all configurations.

The sixth challenge, introduced by enhanced architecture, is SBERT embedding dimensionality management. The 384-dimensional Sentence-BERT output must be concatenated with structured features without the text representation dominating the feature space. This is addressed through PCA reduction to 50 dimensions, fitted exclusively on the training split to prevent data leakage, retaining approximately 80% of embedding variance while producing a balanced 86-dimensional fused matrix.

The seventh challenge is explainability on the hybrid fused model. SHAP's TreeExplainer operates clearly on structured features, but PCA-compressed text embedding dimensions are not directly human-interpretable. This is addressed through a dual approach motivated by Maarouf et al. (CD7): SHAP is applied to the full 86-feature fused matrix with PCA text dimensions labelled as text-derived components, while LIME is separately applied to the raw description text to produce word-level attribution. This ensures both modalities are independently explainable without compromising the interpretability of the structured feature attributions.

8. High-Level Solution Direction

The proposed solution is an upgraded supervised binary classification pipeline with adaptive confidence-based multi-model routing and NLP hybrid fusion, trained on pre-launch Steam game metadata. The pipeline is extended from the original six phases to eight phases, incorporating derived feature engineering and Model E NLP fusion training as new phases motivated by the cross-domain research in Sections 2.2 and 2.3. All phases are scoped for completion within a three-month implementation window, with the optional stacked meta-learner (Priority 5) added only if the first four priorities are complete.

Phase 1: Data Preparation. The Steam Games Dataset 2025 is loaded and cleaned by removing post-launch metrics, URL fields, and non-pre-launch columns. Games with fewer than 100 total reviews are filtered out to ensure label stability. The binary reception label is computed from pct_pos_total using a 75% threshold. The short_description field is cleaned of HTML tags using BeautifulSoup and whitespace normalized. The resulting dataset contains 20,383 games with a 72%/28% class split, of which 20,346 (99.8%) have non-empty clean descriptions.

Phase 2: Derived Feature Engineering. Motivated directly by Sahu et al. (CD1), three interaction features are computed from existing raw inputs: price-to-DLC ratio, genre concentration score, and platform coverage ratio. These are appended to the structured feature matrix before any model training, extending it from 33 to 36 features, and organized into the five graduated tiers described in Section 5.1.

Phase 3: Base Model Evaluation. Four classifiers: Logistic Regression, Random Forest, XGBoost, and CatBoost, are trained and evaluated on the full 36-feature structured matrix under three imbalance handling strategies: no correction, SMOTE, and class weighting. All configurations are evaluated via stratified five-fold cross-validation with F1 score on the minority class as the primary metric. Preliminary results show Random Forest without imbalance correction achieving the strongest F1 of 0.822, establishing the full-feature tabular baseline.

Phase 4: Adaptive Specialist Training and Confidence Router. Four dedicated models are trained in their respective feature tiers: Model A on Core features, Model B on Core and Monetization features, Model C on Core and Content features, and Model D on all 36 structured features. Each specialist uses the best configuration from Phase 3. The confidence-based router is then implemented: at inference time, each eligible specialist produces a predict_proba() output, and the model expressing the highest-class probability is selected. A missing-feature ablation study evaluates performance at each tier against the full-feature baseline.

Phase 5: Model E: NLP Hybrid Fusion Training. The clean game descriptions are encoded using Sentence-BERT (all-MiniLM-L6-v2), the deep learning component of the system, producing 384-dimensional embeddings. PCA fitted on the training split reduces these to 50 dimensions. The reduced embeddings are concatenated with the 36 structured features to produce an 86-dimensional fused matrix, on which Model E is trained. A three-way ablation study is conducted: structured only (Model D) versus TF-IDF fusion versus SBERT early fusion (Model E), directly answering whether developer-written descriptions carry additional pre-launch reception signal.

Phase 6: Dual Explainability Analysis. SHAP is applied to each specialist model (A through E) to generate tier-specific feature attribution analyses, showing which features available at the developer's current planning stage are driving the prediction. Price (importance 0.206) and DLC count (0.082) dominate the full tabular model; the Core-only specialist produces different patterns reflecting what is learnable from genre and price alone. For Model E, LIME is additionally applied to the raw description text, identifying which words in the developer's description most influenced the individual prediction, in line with the explainability blueprint from Maarouf et al. (CD7).

Phase 7: System Deployment. A Streamlit-based risk calculator is built incorporating the confidence-based router and dual explainability. A developer selects their available features, inputs values, and optionally provides a game description. The router queries all eligible specialists, selects the most confident, and returns a risk score, the tier and model used, SHAP attributions for structured features, and when Model E is selected, LIME word attributions for the description. This makes the full system transparent and practically useful to a developer at any planning stage.

Phase 8: Comprehensive Ablation and Evaluation. A systematic ablation study compares performance across all specialist models and routing configurations: rule-based versus confidence-based routing, Model D versus Model E, TF-IDF versus SBERT text encoding, and with versus without derived features. Evaluation metrics across all configurations are F1 (primary), AUC, precision, recall, and cross-validation standard deviation.

Phase	Description	Output	Cross-Domain Motivation	Timeline
1	Data prep: load, clean, filter to 20,383 games, compute binary label, clean short_description HTML	Working dataset; 99.8% description coverage	Original architecture	Week 1–2
2 (new)	Derived feature engineering: price-to-DLC ratio, genre concentration, platform coverage	33 → 36 feature matrix across all tiers	Sahu et al. (CD1): 17.62% accuracy gain from derived features	Week 2–3
3	Base model eval: LR, RF, XGBoost, CatBoost × 3 imbalance strategies on 36-feature matrix	Best config identified; RF no correction F1 0.822 (baseline)	Original architecture	Week 3–5
4	Adaptive specialist training: Models A–D on respective tiers + confidence-based router	4 trained specialists; confidence router; tier ablation results	Router: Alajmi and Pergola (CD11) and Wang et al. (CD12)	Week 5–7
5 (new)	Model E: SBERT encoding → PCA 50 dims → early fusion → 86-dim matrix + 3-way text ablation	Model E trained; Model D vs TF-IDF vs SBERT ablation results	CD6, CD8, CD9 (3 independent domains)	Week 7–10
6 (updated)	Dual explainability: SHAP for all models (A–E) + LIME for text portion of Model E	SHAP tier plots; LIME word attributions for Model E	SHAP: original; LIME addition: Maarouf et al. (CD7)	Week 10–11
7	Streamlit deployment: confidence router + dual XAI interface	Interactive prototype with 5-tier routing and SHAP + LIME display	Original architecture, updated	Week 11–12
8	Comprehensive ablation: routing strategy, text encoding, derived features, tier degradation	Full comparison table across all configurations	Cross-domain research scope	Week 12

The complete pipeline is implemented in Python using scikit-learn, XGBoost, CatBoost, imbalanced-learn, SHAP, lime, sentence-transformers, BeautifulSoup, pandas, and Streamlit, conducted on Google Colab. The implementation and dataset are maintained in a version-controlled repository.

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